

Supplementary Material

Multi-Objective Evolutionary Hyperparameter Optimisation for Recursive Battery Degradation Forecasting

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This document accompanies the main article and records implementation details, supporting tables, additional analyses, and provenance information that are not required to follow the main argument. Numerical values are reproduced from the archived final analysis outputs, and the scope of each supplementary analysis is kept consistent with the main manuscript.

S1 Extended data preparation and feature construction

S1.1 Primary and external datasets

The source-domain analysis used 228 commercial NMC/C–SiO 18650 cells and 3,980 processed health-check records. The external transfer audit used 40 LG M50T 21700 cells and 246 later-half evaluation points. Table S1 records the numerical ranges used in the source-domain analysis.

Table S1: Source-domain data summary.

Property	Value
Cells / processed records	228 / 3,980
Nominal temperatures	0, 10, 25, 40 °C
Charge-rate range	0–1.67 C
Discharge-rate range	0–1 C
Capacity range	1.512790–2.970551 Ah
Electrolyte-resistance range	12.990333–28.150000 mΩ
Beginning-of-life capacity Q_0	2.926306–2.970551 Ah
Beginning-of-life electrolyte resistance $R_{e,0}$	12.990333–19.766333 mΩ
Beginning-of-life pulse-resistance proxy $R_{pulse,0}$	0.020467–11.714533 mΩ
Archived ageing index k_{exp}	0–63,884

The Luh–Blank source dataset is available under CC BY 4.0 (data DOI: [10.35097/1947](https://doi.org/10.35097/1947), version 1). The Kirkaldy LG M50T dataset is available under CC BY 4.0 (Zenodo DOI: [10.5281/zenodo.10637534](https://doi.org/10.5281/zenodo.10637534), version 1).

S1.2 Nine exogenous source-domain features

Table S2 gives the nine exogenous channels used by the recurrent models. Capacity and electrolyte-resistance history were appended separately during sequence construction.

Table S2: Exogenous feature definitions used in the source-domain sequence models.

No.	Feature	Symbol	Definition / status
1	Operating-cycle ageing index	k_{exp}	Cell-wise offset of source field <code>num_cycles_op</code> ; clipped at zero.
2	Temperature	T	Source field <code>age_temp</code> .
3	Beginning-of-life capacity	Q_0	First retained valid capacity record for the cell.
4	Beginning-of-life electrolyte resistance	$R_{e,0}$	First retained valid electrolyte-resistance value.
5	Pulse-resistance proxy	$R_{\text{pulse},0}$	Beginning-of-life difference between 1 s and 10 ms pulse-resistance measurements; not interpreted as fitted charge-transfer resistance.
6	Charge C-rate	C_{chg}	Source field <code>age_chg_rate</code> .
7	Discharge C-rate	C_{dis}	Source field <code>age_dischg_rate</code> .
8	Calendar-ageing SOC setpoint descriptor	S_{SOC}	Source field <code>age_soc</code> .
9	Ageing-type descriptor	A_{type}	Source field <code>age_type</code> : calendar, cyclic, or profile ageing.

The source variable underlying k_{exp} is `num_cycles_op`, defined in the official data dictionary as the number of cycles performed under the applicable operating-ageing conditions specified by ageing type, temperature, SOC, charge rate, discharge rate, and profile. In preprocessing it was transformed separately for each cell as

$$k_{\text{exp},i} = \max(\text{num_cycles_op}_i - \text{num_cycles_op}_{\text{first}}, 0). \quad (\text{S1})$$

Because the first retained value was normally zero, the archived processed k_{exp} and source operating-cycle count were generally identical. This variable is not an equivalent-full-cycle count, cumulative-throughput measure, elapsed-time measure, or checkup index; equivalent full cycles are defined separately in the source data structure from charged and discharged ampere-hour throughput.

The source SOC variable is `age_soc`. Its values 10, 50, 90, and 100 denote calendar-ageing SOC setpoints of 10%, 50%, 90%, and 100%, respectively. A value of 0 denotes that a calendar-ageing SOC setpoint is not applicable, as in cyclic and profile ageing. The archived processed column name `soc_window` is therefore retained only as an implementation label; the manuscript describes this channel as the calendar-ageing SOC setpoint descriptor.

The source ageing-type variable is `age_type`. The top-level codes are 1 for calendar ageing, 2 for cyclic ageing, and 3 for profile ageing. Profile subtypes, including complete WLTP and extra-high WLTP protocols, are distinct lower-level protocol codes and are not substituted for these top-level values.

In the archived 3,980-row table, the `age_soc` values 0, 10, 50, 90, and 100 occurred 2,606, 348, 348, 342, and 336 times, respectively. The top-level `age_type` codes 1, 2, and 3 occurred 1,374, 1,837, and 769 times. The source-field definitions above are documented in `Data Structure v05.3.xlsx`; the corrected preprocessing branch is preserved in `pinn-battery (27).ipynb` and in the repository script identified below.

S1.3 Documented preprocessing operations

The preserved data-construction code sorted records within cell by the source ageing record and timestamp, retained the latest timestamp when duplicate ageing records were present, and rejected capacity values at or below 0.1 Ah. Electrolyte resistance R_e was taken from the available reference-impedance measurements; where those values were absent and pulse data were available, the 10 ms pulse-resistance value was used as a fallback. Missing resistance values within otherwise valid trajectories were interpolated in both directions.

The pulse-timescale resistance proxy was constructed as

$$R_{\text{pulse}}(k) = R_{1s}(k) - R_{10ms}(k). \quad (\text{S2})$$

The beginning-of-life value was the first non-missing proxy after interpolation. For cells without a valid pulse-resistance difference, the documented fallback was

$$R_{\text{pulse},0} = 0.35R_{e,0}. \quad (\text{S3})$$

This channel is therefore a pulse-timescale resistance difference and is not a fitted electrochemical R_{ct} quantity. Feature scaling was fitted on training cells only. For feature j ,

$$\tilde{x}_{i,j} = \frac{x_{i,j} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}}}. \quad (\text{S4})$$

The fitted transformation was then applied unchanged to validation and held-out cells. Target scaling followed the same training-only rule, and all reported prediction errors were calculated after inverse transformation.

A complete corrected preprocessing implementation is preserved in the public repository. The script is available at

[preprocess_phase2_split.py](#).

It corresponds to commit `f80ac44`. The script reproduces the corrected preprocessing logic and the legacy 180/27/21-cell PINN split. All 13 repository tests passed for this repository state.

The corrected preprocessing implementation reads 240 EOC files and the corresponding EIS files; extracts capacity from `cap_aged_est_Ah`; extracts R_e from `z_ref_now_mOhm`; retains `z_ref_init_mOhm` in the legacy resistance branch; maps `age_temp`, `age_chg_rate`, `age_dischg_rate`, `age_soc`, `num_cycles_op`, and `age_type`; removes invalid or non-positive targets; applies the protocol filters; enforces non-increasing capacity and non-decreasing resistance trajectories; constructs Q_0 , $R_{e,0}$, $R_{\text{pulse},0}$, and k_{exp} ; performs a seeded cell-level split; and writes `phase2_train.csv`, `phase2_val.csv`, and `phase2_test.csv`.

Its recorded corrected execution produced 180 training cells and 3,157 rows, 27 validation cells and 392 rows, and 21 test cells and 395 rows, for a total of 228 cells and 3,944 rows. The archived combined table used by the final optimisation study contains 3,980 rows. The 3,944-row execution is therefore documented as a corrected preprocessing revision and legacy PINN split, while the final optimisation results in the main manuscript remain tied to the archived 3,980-row table.

S1.4 Cell-level partitions

The final optimisation-stage source-domain partition contained 209 training cells (3,461 rows), 9 validation cells (243 rows), and 10 held-out test cells (276 rows). The common recursive scoring window began at trajectory row index 20 and contained 63 validation points and 76 test points. No sequence crossed a cell boundary.

S2 Candidate LSTM and training implementation

S2.1 Forecasting network

Each recurrent step contained 11 channels: nine exogenous variables plus the preceding capacity and electrolyte-resistance values. Candidate-dependent LSTM depth and hidden width were followed by a fixed regression head,

$$\mathbf{h}_{1:L} = \text{LSTM}_{\theta}(\mathbf{u}_{1:L}), \quad (\text{S5})$$

$$\mathbf{z}_1 = \text{ReLU}(W_1 \mathbf{h}_L + \mathbf{b}_1), \quad (\text{S6})$$

$$\mathbf{z}_2 = \text{ReLU}(W_2 \text{Dropout}_{0.1}(\mathbf{z}_1) + \mathbf{b}_2), \quad (\text{S7})$$

$$\hat{\mathbf{y}} = W_3 \mathbf{z}_2 + \mathbf{b}_3, \quad (\text{S8})$$

with dimensions $d_h \rightarrow 128 \rightarrow 64 \rightarrow 2$. LSTM dropout was 0.2 when recurrent depth exceeded one and zero for one-layer candidates.

S2.2 Teacher-forced pretraining

Teacher-forced pretraining used AdamW with candidate learning rate η , weight decay 10^{-5} , batch size 512, Smooth-L1 prediction loss in scaled target space, and gradient-norm clipping at 1.0. Validation macro MAPE controlled early stopping, and the best checkpoint was restored before recursive fine-tuning.

S2.3 Recursive fine-tuning

Each rollout segment contained an initial history of length L , followed by H future exogenous rows and H target rows. Rollout segments were generated with stride 2. At each future step, predicted capacity and resistance were concatenated with the known exogenous variables and returned to the next recurrent input. The rollout loss was

$$\mathcal{L}_{\text{RO}} = \frac{1}{H} \sum_{h=1}^H \|\hat{\mathbf{y}}_{k+h} - \mathbf{y}_{k+h}\|_2^2. \quad (\text{S9})$$

AdamW was used at learning rate 0.30η , weight decay 10^{-5} , batch size 128, and gradient-norm clip 1.0. Early stopping was based on common-window validation rollout macro MAPE.

S2.4 Stage-specific schedules

Table S3: Maximum epochs and early-stopping patience used at each stage.

Stage	Teacher max.	Teacher patience	Rollout max.	Rollout patience
Search	80	18	15	5
Re-screening	160	30	30	10
Final confirmation	220	40	40	15

S3 Evolutionary and random-search implementation

S3.1 Search representation and feasibility

Candidates were represented as $[L, d_h, n_{\text{LSTM}}, H, \eta]$. The discrete domains were $L \in \{10, 15, 20\}$, $d_h \in \{64, 96, 128, 160, 192\}$, $n_{\text{LSTM}} \in \{1, 2, 3\}$, and $H \in \{3, 5, 8, 10, 15\}$. Learning rate was sampled in logarithmic space over 10^{-4} to 2×10^{-3} . Feasibility required $L + H \leq 29$.

S3.2 Crossover, mutation, repair, and duplicate handling

Uniform crossover was applied field-wise. If parent vectors were $\omega^{(1)}$ and $\omega^{(2)}$, each offspring field j was inherited independently as

$$\omega_j^{(c)} = \begin{cases} \omega_j^{(1)}, & U_j < 0.5, \\ \omega_j^{(2)}, & U_j \geq 0.5, \end{cases} \quad U_j \sim \mathcal{U}(0, 1). \quad (\text{S10})$$

Each discrete field was mutated independently with probability 0.25. Learning rate was mutated with probability 0.25 in $\log_{10}$ space,

$$\log_{10} \eta' = \text{clip} \left(\log_{10} \eta + \epsilon, \log_{10} 10^{-4}, \log_{10} (2 \times 10^{-3}) \right), \quad \epsilon \sim \mathcal{N}(0, 0.28^2). \quad (\text{S11})$$

If no field changed, one randomly selected field was forced to mutate. For infeasible offspring, the generated L was retained and H was replaced by the closest allowed value satisfying $L + H \leq 29$; ties favoured the larger feasible H . Duplicate candidates were regenerated and did not consume the unique-evaluation budget.

S3.3 NSGA-II and NSGA-III

NSGA-II used binary tournament selection based first on Pareto rank and then crowding distance. Parent and offspring populations were combined, complete nondominated fronts were retained, and a partially accepted final front was truncated by descending crowding distance.

NSGA-III used the same candidate representation, variation operators, feasibility handling, population size, and candidate evaluator. For the three-objective problem, a Das–Dennis construction with division parameter $p = 6$ produced 28 reference directions, matching the population size. Objectives were min–max normalised within the combined population for reference-direction association. Split-front filling used the least-occupied niches, selecting the closest associated point for an empty niche and a random associated point otherwise.

S3.4 Matched random search

Random search sampled the same four discrete domains uniformly and sampled learning rate uniformly in log space. It used the same feasibility rule, training schedule, objective definitions, latency routine, run-wise training seeds, and 280-unique-evaluation budget.

S3.5 Search-budget integrity audit

Each method was run 15 times, with population/reference size 28 and 280 unique candidate evaluations per run. Run RNG seeds 20270001–20270015 and candidate-training seeds 41001–41015 were paired by run index across methods. The archived audit recorded 28 initial candidates for every run and no missing objectives, duplicate evaluation identifiers, or infeasible retained candidates.

The initial-population audit additionally found exact agreement of candidate-ID sequences and canonical little-endian float64 encodings of $[L, d_h, n_{\text{LSTM}}, H, \eta]$ across the 45 matched method-run comparisons against the corresponding NSGA-II run. Training seed was constant within each method-run initial population. These checks support the matched-initialisation design; later search trajectories were method specific.

S4 Pareto indicators and statistical calculations

S4.1 Metric definitions

For target j with N observations,

$$\text{RMSE}_j = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{i,j} - \hat{y}_{i,j})^2}, \quad (\text{S12})$$

$$R_j^2 = 1 - \frac{\sum_{i=1}^N (y_{i,j} - \hat{y}_{i,j})^2}{\sum_{i=1}^N (y_{i,j} - \bar{y}_j)^2}. \quad (\text{S13})$$

Macro MAPE was the arithmetic mean of capacity MAPE and electrolyte-resistance MAPE. Seed uncertainty was summarised using sample standard deviation,

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (\text{S14})$$

S4.2 Common transformed objective space

All 12,600 search evaluations were transformed to

$$\mathbf{g} = [\text{MAPE}_{\text{macro}}, \log_{10} P, \log_{10} T]. \quad (\text{S15})$$

The global lower and upper bounds were

$$\mathbf{g}_{\min} = [0.7308796223, 4.5613160916, -0.4342352697], \quad (\text{S16})$$

$$\mathbf{g}_{\max} = [8.8913714513, 5.8939979806, 0.0246944637]. \quad (\text{S17})$$

Normalisation used

$$\tilde{g}_j = \frac{g_j - g_{\min,j}}{g_{\max,j} - g_{\min,j}}. \quad (\text{S18})$$

The same bounds were held fixed for every method, run, and checkpoint.

S4.3 Hypervolume, IGD, IGD⁺, and spacing

Hypervolume was calculated in the common normalised minimisation space using the reference point (1.1, 1.1, 1.1). The pooled IGD reference set was the nondominated union of all 12,600 evaluations and contained 28 points.

For approximation set A and reference set R , conventional IGD was

$$\text{IGD}(A, R) = \frac{1}{|R|} \sum_{\mathbf{r} \in R} \min_{\mathbf{a} \in A} \|\mathbf{a} - \mathbf{r}\|_2. \quad (\text{S19})$$

The dominance-aware IGD⁺ distance from approximation point $\mathbf{a}$ to reference point $\mathbf{r}$ was

$$d^+(\mathbf{a}, \mathbf{r}) = \sqrt{\sum_j \max(a_j - r_j, 0)^2}, \quad (\text{S20})$$

and IGD⁺ replaced the Euclidean distance in Eq. (S19) by d^+ . Leave-one-method-out IGD rebuilt the pooled reference set after excluding all evaluations from the method being scored; each resulting reference set contained 46 points.

For a nondominated set of size M , Schott spacing was based on nearest-neighbour Manhattan distances d_i ,

$$S = \sqrt{\frac{1}{M-1} \sum_{i=1}^M (d_i - \bar{d})^2}. \quad (\text{S21})$$

Smaller spacing indicates more even local separation. Indicators were recomputed at 28, 56, ..., 280 evaluations.

S4.4 Inference procedures

Optimiser-level omnibus comparisons used Kruskal–Wallis tests. Pairwise follow-up comparisons used two-sided Mann–Whitney U tests with Holm correction. Vargha–Delaney A_{12} was reported as the probability that an indicator value from the first-listed method exceeded one from the second-listed method. For minimised indicators, $A_{12} < 0.5$ favours the first-listed method.

Final-model seed comparisons used paired two-sided Wilcoxon signed-rank tests. Confidence intervals for mean performance and paired seed differences used 10,000 bootstrap resamples at the seed level. External-transfer confidence intervals used 10,000 bootstrap resamples of the 40 paired cell effects.

S5 Training-noise characterisation

Eight fixed configurations were retrained over ten seeds under the search schedule. Table S4 gives the complete results. The macro-MAPE SD ranged from 0.212649 to 0.516392 percentage points, while latency SD remained below 0.03 ms.

Table S4: Fixed-configuration training-noise experiment. MAPE values are percentage points; latency is in milliseconds.

Configuration	MAPE mean	MAPE SD	Latency mean	Latency SD
L10_D128_N2_H15_LR0.00070000	1.955233	0.454058	0.419271	0.006028
L10_D192_N3_H10_LR0.00100000	1.550062	0.259948	0.741568	0.003483
L10_D64_N1_H3_LR0.00070000	3.263282	0.464884	0.399583	0.012836
L15_D64_N2_H10_LR0.00120000	1.754318	0.261980	0.426386	0.012103
L15_D96_N1_H5_LR0.00051649	2.065986	0.212649	0.406729	0.026333
L20_D160_N3_H5_LR0.00020000	2.436634	0.261370	0.705775	0.001236
L20_D192_N2_H8_LR0.00100000	1.828346	0.516392	0.702592	0.003302
L20_D96_N1_H3_LR0.00042092	2.665071	0.224614	0.414777	0.013353

The experiment was descriptive. None of these configurations was promoted into the final comparison because of its noise-floor outcome.

S6 Extended optimisation results

S6.1 Primary and alternate Pareto indicators

Table S5: Final indicator values after 280 evaluations. Values are mean $\pm$ SD across 15 independent runs.

Method	HV	IGD	IGD ⁺	LOO-IGD
NSGA-II	1.197236 $\pm$ 0.018458	0.070110 $\pm$ 0.011361	0.040469 $\pm$ 0.010461	0.058144 $\pm$ 0.017707
NSGA-III	1.157911 $\pm$ 0.024419	0.080153 $\pm$ 0.014388	0.059506 $\pm$ 0.012560	0.081232 $\pm$ 0.014767
Random search	1.141019 $\pm$ 0.016368	0.092198 $\pm$ 0.013190	0.071512 $\pm$ 0.009126	0.093927 $\pm$ 0.019054

Table S6: Optimiser-level statistical comparisons. Pairwise p -values are Holm corrected. A_{12} is the probability that the first-listed method has the larger indicator value.

Metric	Comparison	p	A_{12}
HV	Kruskal–Wallis, all methods	2.112×10^{-6}	–
HV	NSGA-II vs NSGA-III	2.713×10^{-4}	0.9111
HV	NSGA-II vs random search	1.243×10^{-5}	0.9956
HV	NSGA-III vs random search	0.0890	0.6844
IGD	Kruskal–Wallis, all methods	3.658×10^{-4}	–
IGD	NSGA-II vs NSGA-III	0.0559	0.2622
IGD	NSGA-II vs random search	4.069×10^{-4}	0.0889
IGD	NSGA-III vs random search	0.0559	0.2889
IGD ⁺	Kruskal–Wallis, all methods	1.626×10^{-6}	–
IGD ⁺	NSGA-II vs NSGA-III	5.243×10^{-4}	0.1067
IGD ⁺	NSGA-II vs random search	1.243×10^{-5}	0.0044
IGD ⁺	NSGA-III vs random search	0.0225	0.2533
LOO-IGD	Kruskal–Wallis, all methods	1.924×10^{-5}	–
LOO-IGD	NSGA-II vs NSGA-III	9.874×10^{-4}	–
LOO-IGD	NSGA-II vs random search	1.206×10^{-4}	–
LOO-IGD	NSGA-III vs random search	0.0680	–

S6.2 Front size and spacing

Table S7: Final front cardinality and Schott spacing.

Method	Front size, mean $\pm$ SD	Spacing, mean $\pm$ SD
NSGA-II	20.733 $\pm$ 5.391	0.084018 $\pm$ 0.061764
NSGA-III	21.133 $\pm$ 4.565	0.082378 $\pm$ 0.049864
Random search	21.467 $\pm$ 4.340	0.070273 $\pm$ 0.043700

The front-size omnibus test gave $H = 0.137$, $p = 0.934$, and the spacing test gave $H = 0.040$, $p = 0.980$. All pairwise Holm-corrected p -values for these two outcomes were 1.0. The pooled nondominated union of all 12,600 evaluations contained 28 points: 21 from NSGA-II, 5 from random search, and 2 from NSGA-III.

S6.3 Per-run minimum validation MAPE

Table S8: Distribution of the smallest validation macro MAPE observed within each 280-evaluation run.

Method	Mean	SD	Median	Min	Max
NSGA-II	0.951754	0.075866	0.963294	0.839773	1.114710
NSGA-III	0.948624	0.089330	0.925628	0.839773	1.122576
Random search	0.963720	0.107559	0.970694	0.730880	1.135640

No exact candidate-ID match was found between the 45 run winners and the independently retrained noise-floor or re-screening panels. A paired winner’s-curse estimate was therefore not computed, and no approximate candidate matching was substituted.

S6.4 Final configuration, compute, and seed dispersion

Table S9: Ten-seed final configuration and accuracy summary.

Configuration	L	Hidden	Layers	H	Learning rate	Macro MAPE $\pm$ SD
Manual	20	192	2	8	0.001000000	1.855862 $\pm$ 0.226975
NSGA-II	15	64	1	8	0.001215422	1.809732 $\pm$ 0.371581
NSGA-III	20	64	1	8	0.000868730	1.871584 $\pm$ 0.349391
Random search	15	64	1	5	0.000739324	1.862183 $\pm$ 0.155434

Relative to the manual configuration, every optimiser-selected finalist reduced parameter count by 92.521633%. The NSGA-II, NSGA-III, and random-search forward-latency reductions were 38.925%, 39.339%, and 40.619%; recorded final-training-time reductions were 62.222%, 71.544%, and 60.803%; trajectory-latency reductions were 35.308%, 38.042%, and 36.192%, respectively. For macro MAPE, all three paired comparisons against the manual reference had Holm-corrected $p = 1.0$.

The NSGA-II minus manual macro-MAPE difference was -0.046130 percentage points with 95% CI $[-0.383561, 0.229344]$ and raw $p = 0.7695$. Levene’s and Brown–Forsythe tests for seed-level variance gave $p = 0.1316$ and $p = 0.3134$, respectively; numerical SD differences were therefore not treated as established variance effects.

S6.5 Post-search selection sensitivity

Table S10: Sensitivity of finalist choice to the post-search decision rule. Re-screened errors use three seeds.

Method	Minimum re-screened MAPE	Minimum MAPE with $P \leq 100k$	Implemented knee
NSGA-II	L15_D64_N1_H8, 2.3275 ± 0.4095 , 36,418 params	Same	Same
NSGA-III	L15_D96_N1_H3, 1.9489 ± 0.1416 , 62,658 params	Same	L20_D64_N1_H8, 2.2853 ± 0.0983 , 36,418 params
Random search	L15_D64_N1_H5, 2.4432 ± 0.4458 , 36,418 params	Same	Same

The larger NSGA-III accuracy-priority candidate was not evaluated in the ten-seed final-confirmation stage; its result remains a three-seed re-screening observation.

S7 Auxiliary sequence and physics-guided models

S7.1 TCN baseline

The TCN received the same 11 per-step channels as the LSTM. An input projection convolution mapped 11 channels to 96 channels with kernel size 1. Four causal residual blocks used dilations 1, 2, 4, and 8. Each block contained a kernel-3 causal convolution with causal trimming, batch normalisation, ReLU, dropout 0.1, and residual addition. The output head was $96 \rightarrow 64 \rightarrow 2$ with ReLU and dropout 0.1. The model contained 119,234 trainable parameters and used a sequence length of 20.

The TCN was rerun on an NVIDIA Tesla T4 using Python 3.12.13, PyTorch 2.10.0+cu128, CUDA 12.8, and cuDNN 9.10.2. Training used a learning rate of 7×10^{-4} , weight decay 10^{-4} , batch size 512, a maximum of 180 epochs, and early-stopping patience 35. Seeds were 42, 52, 62, 72, and 82. Recursive evaluation used the same common held-out window of 76 observations from all 10 test-cell trajectories.

Table S11: Aggregate TCN results on the NVIDIA Tesla T4. Values are mean $\pm$ standard deviation across the five training seeds unless stated otherwise.

Metric	Result
Macro MAPE	$1.977483 \pm 0.357416\%$
Capacity MAPE	$3.019229 \pm 0.489898\%$
Electrolyte-resistance MAPE	$0.935737 \pm 0.402116\%$
Capacity RMSE	0.129352 ± 0.024427 Ah
Electrolyte-resistance RMSE	0.224971 ± 0.070272
Capacity R^2	0.768000 ± 0.080710
Electrolyte-resistance R^2	0.981119 ± 0.011443
Recorded training time	2.210750 ± 0.664980 s per seed
Trajectory latency	12.558584 ± 0.033061 ms
Completed epochs	105.8 ± 25.9
Trainable parameters	119,234
Evaluation points / cells	76 / 10

The trajectory-latency uncertainty in Table S11 is the standard deviation across the five seed-level latency means. Each seed used five warm-up recursive rollouts followed by 100 synchronized timed rollouts. Across all 500 individual measurements, the pooled latency was 12.558584 ± 0.428708 ms per trajectory, with median 12.476176 ms and range 11.813613–15.501314 ms.

Table S12: Per-seed TCN predictive results on the common 76-point held-out recursive window.

Seed	Macro MAPE (%)	Q MAPE (%)	R_e MAPE (%)	Q RMSE (Ah)	R_e RMSE	$Q R^2$	$R_e R^2$
42	1.993337	3.367407	0.619267	0.153091	0.166311	0.684045	0.990429
52	2.335297	3.085300	1.585294	0.116954	0.322895	0.815603	0.963921
62	2.188475	3.440013	0.936937	0.143271	0.227581	0.723277	0.982077
72	1.395954	2.208834	0.583074	0.092673	0.150001	0.884219	0.992214
82	1.974352	2.994590	0.954115	0.140770	0.258069	0.732856	0.976954

Table S13: Per-seed TCN training and synchronized recursive-trajectory timing. Each latency value is mean $\pm$ standard deviation across 100 repeated measurements after five warm-up rollouts.

Seed	Training time (s)	Latency (ms/trajectory)	Epochs
42	3.204896	12.541497 ± 0.501344	109
52	1.594276	12.518410 ± 0.320943	78
62	2.555570	12.547600 ± 0.484610	146
72	1.950818	12.589807 ± 0.369261	107
82	1.748190	12.595605 ± 0.443184	89

The synchronized end-to-end latency measurement included input scaling, host-to-device transfer, batch-one TCN forward passes, device-to-host transfer, inverse scaling, output clipping, and recursive feedback through the complete trajectory. Dataset loading, model loading, and metric calculation were excluded from the timed region.

S7.2 Mamba baseline

The Mamba baseline used the official `mamba_ssm`.Mamba implementation from `mamba-ssm 2.3.2.post1` under Python 3.12.13 and PyTorch 2.10.0+cu128 on an NVIDIA Tesla T4 with CUDA. The package was installed using `pip install mamba-ssm --no-build-isolation` and retained the default `use_fast_path=True` setting.

An input projection mapped the 11 sequence channels to `d_model=64`, followed by two residual Mamba blocks. Each block used `d_state=16`, `d_conv=4`, and `expand=2`. A final normalisation and two-output prediction head used the final sequence position. The model contained 70,722 trainable parameters. Sequence length was 15, common evaluation started at index 20, and rollout horizon was 10.

Training used AdamW at learning rate 7×10^{-4} with weight decay 10^{-5} , teacher batch size 256, rollout batch size 128, 220 teacher-forced epochs, and 40 rollout fine-tuning epochs. Seeds were 42, 52, 62, 72, and 82. Across these five seeds, macro MAPE was $2.945251 \pm 0.208178\%$, capacity MAPE $4.392401 \pm 0.471298\%$, and electrolyte-resistance MAPE $1.498101 \pm 0.166333\%$. Recorded training time was 12.038 s and trajectory latency 13.925 ms on the Tesla T4.

S7.3 Physics-informed baseline

The capacity-residual term used

$$\frac{d\hat{Q}}{dk} = -\alpha_0 \exp\left(-\frac{E_a}{RT}\right) \hat{Q}^\beta, \quad (\text{S22})$$

with residual

$$\mathcal{R}_Q = \frac{d\hat{Q}}{dk} + \alpha_0 \exp\left(-\frac{E_a}{RT}\right) \hat{Q}^\beta. \quad (\text{S23})$$

The resistance residual was

$$\mathcal{R}_{R_e} = \frac{d\hat{R}_e}{dk} - g_\theta(k, T, \mathbf{x}). \quad (\text{S24})$$

The full training loss was

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{data}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} + \gamma r(e) \left[\lambda_m \mathcal{L}_{\text{mono}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_p \mathcal{L}_{\text{phys}} \right], \quad (\text{S25})$$

where

$$\mathcal{L}_{\text{phys}} = \|\mathcal{R}_Q\|_2^2 + 0.3 \|\mathcal{R}_{R_e}\|_2^2 + \mathcal{L}_{\text{diversity/prior}}. \quad (\text{S26})$$

Adaptive gradient-norm balancing initialised the coefficients at 1, updated them every five epochs, and used EMA momentum 0.8. Allowed ranges were $\lambda_m \in [0.01, 5]$, $\lambda_s \in [0.001, 1]$, $\lambda_p \in [0.02, 0.5]$, and $\lambda_{\text{IC}} \in [0.01, 3]$. The activation-energy prior was 56 kJ mol⁻¹ with scale 7, a diversity floor of 4, and allowed range 40–80 kJ mol⁻¹.

The data-only backbone obtained 7.560 ± 0.195% macro MAPE, while the physics-constrained version obtained 8.049 ± 0.294%. The recovered full-model activation energy was approximately 57.5 kJ mol⁻¹. Across the 180 phase-one training-cell fits, activation energy ranged from 51.75 to 61.48 kJ mol⁻¹ with mean 55.69 ± 2.86 kJ mol⁻¹. These quantities are prior-guided fitted parameters, not independent physical measurements.

S7.4 Arrhenius-derived stress ablation

The activation-energy distribution used for this auxiliary descriptor was estimated in the separate phase-one branch using a fixed cell-level split of 180 training, 27 validation, and 21 test cells. Cell-specific E_a values were fitted from the 180 training cells only. Validation and test cells did not contribute to estimation of the activation-energy distribution. For a non-training or external cell without a cell-specific estimate, the global mean of the 180 training-cell estimates was used.

The added descriptor was

$$S_{\text{Arr}} = \ln[\max(k_{\text{exp}}, 10^{-6})] - \frac{E_a}{RT}, \quad (\text{S27})$$

with $R = 8.314 \times 10^{-3}$ kJ mol⁻¹ K⁻¹. The larger model was initialised first; the matched model without the stress channel was obtained by deleting that input column and copying all same-shaped parameters exactly.

Table S14: Matched-seed stress-descriptor ablation. Values are mean ± SD across ten seeds.

Variant	Macro MAPE	Q MAPE	R_e MAPE
Base four-feature representation	2.587153 ± 0.341377	3.360104 ± 0.435605	1.814202 ± 0.641280
Base + S_{Arr}	3.209851 ± 0.434305	3.655341 ± 0.469357	2.764361 ± 0.507849

The mean deterioration was 0.622698 percentage points; macro MAPE worsened on all ten matched seeds and the paired Wilcoxon test gave $p = 0.001953$. Two implementation properties are relevant when interpreting this result. First, k_{exp} and T were already direct input channels, while fitted E_a varied over a relatively narrow range. Second, the archived 3,980-row table contained 1,554 rows with $k_{\text{exp}} = 0$, consisting primarily of calendar-ageing observations for which the operating-cycle count remains zero together with initial cyclic/profile records. Clipping those values to 10^{-6} maps the logarithmic term to $\ln(10^{-6}) = -13.82$, whereas the corresponding contribution at $k_{\text{exp}} = 1$ is zero. The descriptor therefore introduces a 13.82-unit discontinuity between zero and one operating cycle. The fitted E_a quantity remains a model-derived input rather than an independently measured physical variable, so this result is interpreted as an auxiliary sensitivity analysis.

S8 Grouped five-fold cell-level analysis

The outer split used `GroupKFold(n_splits=5)` on sorted unique cell identifiers, without shuffling or stratification. Within each outer-training set, 15% of cells were assigned to validation after fold-specific shuffling with `default_rng(42+fold)`. Table S15 gives the resulting partition sizes.

Table S15: Grouped five-fold cell and row counts.

Fold	Train cells	Val cells	Held-out cells	Train rows	Val rows	Held-out rows
1	155	27	46	2,555	579	846
2	155	27	46	2,760	405	815
3	155	27	46	2,773	390	817
4	156	27	45	2,818	416	746
5	156	27	45	2,666	558	756

The PINN-derived stress representations used the phase-one training-derived E_a values described above; E_a was not re-estimated independently within each grouped outer fold. These variants therefore evaluate the archived representation under new cell partitions rather than a fold-specific re-fit of the physical parameter.

Seven predefined variants were ranked within each held-out fold. Their mean ranks were 1.40 for the teacher-forced LSTM, 2.60 for the recursive-rollout LSTM, 3.20 for the PINN-derived (k , $R_{e,0}$, Q_0 , S_{Arr}) feature model, 3.60 for the sparse LSTM, 4.20 for the corresponding feature model augmented with $R_{pulse,0}$, 6.00 for the data-only PINN, and 7.00 for the physics-constrained PINN.

The Friedman test gave $\chi^2 = 24.171429$ and $p = 4.85663 \times 10^{-4}$. The Nemenyi critical difference at $\alpha = 0.05$ was 4.028172 ranks. With five folds, the minimum non-zero exact two-sided Wilcoxon p -value is 0.0625, so pairwise superiority was not inferred. The outer folds were not stratified by temperature, C-rate, or ageing protocol.

S9 External transfer implementation

S9.1 Trajectory split

For a target trajectory containing n ordered records, the calibration cutoff was

$$k_{\text{cut}} = \min(\max[1, \lfloor (n - 1) \times 0.50 \rfloor], n - 2). \quad (\text{S28})$$

Calibration rows were indices $0, \dots, k_{\text{cut}}$, training targets used indices $1, \dots, k_{\text{cut}}$, and evaluation began at $k_{\text{cut}} + 1$ and continued to the final record. The split was by ordered record position rather than by a numerical threshold on k_{exp} .

S9.2 Adaptation conditions

The principal transfer condition froze the source-trained representation and adapted only the prediction head using target calibration data. The target-only neural control was trained from scratch using the same target-domain information. A deterministic square-root-fade model was retained as an analytical reference. Full-network fine-tuning, partial unfreezing, and explicit domain-adaptation methods were not evaluated.

Only capacity was used for cross-dataset inference because source-domain electrolyte resistance and the external 0.1 s pulse-resistance quantity are not physically interchangeable.

S9.3 Supported paired comparisons

Table S16: Cell-level external-transfer comparisons on 40 LG M50T cells. Positive differences indicate larger MAPE after source-representation reuse.

Comparison	Mean difference (pp)	95% CI (pp)	Wilcoxon result
Source-pretrained head adaptation – target-only neural model	+6.987501	[5.187332, 8.794860]	$W = 25, p = 1.6 \times 10^{-9}$
Source-pretrained head adaptation – square-root-fade model	+6.772300	[4.906669, 8.808414]	$W = 16, p < 10^{-9}$

The median paired difference relative to target-only training was +5.034458 percentage points.

S10 Reproducibility information

S10.1 Software and hardware environments

Table S17: Execution environments recorded for the principal experiment groups.

Experiment group	Recorded environment
Primary optimiser study	Kaggle Notebook; Linux; NVIDIA Tesla T4; CUDA; Python 3.12.13; PyTorch 2.10.0+cu128.
TCN baseline	Kaggle Notebook; NVIDIA Tesla T4; Python 3.12.13; PyTorch 2.10.0+cu128; CUDA 12.8; cuDNN 9.10.2.
Mamba baseline	Kaggle Notebook; NVIDIA Tesla T4; CUDA; Python 3.12.13; PyTorch 2.10.0+cu128; mamba-ssm 2.3.2.post1; package default <code>use_fast_path=True</code> .
Round-two analyses	Kaggle Notebook; Linux runtime with Tesla T4 available; analysis scripts selected <code>device="cpu"</code> ; Python 3.14.3; NumPy 2.4.2; pandas 3.0.3; PyTorch 2.12.0+cpu; SciPy 1.17.1; scikit-learn 1.8.0; Matplotlib 3.10.9. PyTorch CPU execution was capped at 8 threads, with one CPU thread used for inference-latency measurement.

S10.2 Recorded model-training time

Table S18: Recorded model-training components.

Component	Recorded training time (s)
NSGA-II search	21,851.636600
NSGA-III search	22,068.587345
Random search	23,705.430540
Noise-floor experiment	461.038679
Re-screening	208.372196
Final confirmation	338.598664
Total	68,633.324024 (19.064907 h)

The preserved search wall-clock measurements were 6.154517 h for NSGA-II and 6.217348 h for NSGA-III; these timers excluded the optional preceding noise-floor stage.

S10.3 Repository, archived release, and corrected preprocessing script

The public repository associated with the study is

<https://github.com/Simhaatt/battery-moo-recursive-forecasting>.

The archived release is available at

<https://doi.org/10.5281/zenodo.22057384>.

The corrected preprocessing and legacy 180/27/21-cell PINN split are implemented in

`scripts/preprocess_phase2_split.py`,

at repository commit `f80ac44`. This script reproduces the corrected 3,944-row preprocessing execution described in Section S1 and writes the corresponding phase-two train, validation, and test tables. All 13 repository tests passed for this repository state.

The final optimisation study reported in the main manuscript remains based on the archived 3,980-row processed table and its 209/9/10-cell split. The corrected 3,944-row execution is retained to document the preprocessing logic and the separate legacy PINN branch rather than to replace the data artifact from which the final optimisation results were computed.

End of Supplementary Material